AOU
الجامعة العربية المفتوحة
Arab Open University

Faculty of Business Administration

FIN342
Financial & Securities Markets
Version A

Mid-term Examination
2024-2025 / Spring

22-April-2025

AOU EXAM

Number of Exam Pages: 3 (including this cover sheet)	Time Allowed: 2 hours

Instructions:

1- Take **FIVE (5)** minutes to read all instructions carefully before attempting to write.
2- Mobile phones must be in "SWITCHED **OFF**" mode throughout the examination.
3- This is a **CLOSED** book examination. NO documents of any kind are allowed.
4- You **MUST** answer the questions using only the answer booklets provided
5- You **MUST** return the examination paper together with your answer booklets.
6- Do **NOT** write your answers in pencil except graphs/sketches
7- Students are **NOT** allowed to use any type of dictionary or mobile phones during the examinatior
8- Students are allowed to use ***non-programmable*** calculators only (if needed).
9- Any form of cheating will **NOT** be tolerated and will subject you to AOU Cheating Policy.

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Solve All questions

Question 1: (10 marks) (30 min)

Government economists have forecasted one-year T-bill rates for the following five years as follows:

Year	1-year rate
1	4.25%
2	5.15%
3	5.50%
4	6.25%
5	7.10%

You have liquidity premium 0.25% for the next two years and 0.50% thereafter. Would you be willing to purchase a 4-year T-bond at a 5.75% interest rate?

Question 2: (10 marks) (20 min)

Suppose you want to earn an annualized discount rate of 2.5%. What would be the most you would pay for a 190-day Treasury bill that pays $ 9,000 at maturity?

Question 3: (12 marks) (15 min)

The one-year interest rate over the next 10 years will be 5%, 6.5%, 7%, 8.5%, 10%, 11.5%, 15%, 16%, 17%, 18.5%. Using the pure expectations theory, what will be the interest rates on a 3-year bond, 6-year bond, and 9-year bond?

Question 4: (11 marks) (20 min)

XYZ bond promises the below cash flows. The required discount rate is 10%.

Year	**0**	**1**	**2**	**3**	**4**
Promised Payments		**150**	**180**	**210**	**240**

You plan to buy this bond, hold it for 3½ years, and then sell the bond.

a. Calculate the total cash that you will earn from the bond after the 3½ years? Assuming that periodic cash flows are reinvested at 10%.
b. After buying ABC bond, all market interest rates drop to 9% (including your reinvestment rate), what will be the impact on your total cash flow after 3½ years? How does this compare to part (a)

Question 5: (7 marks) (5 min)

Some economists suspect that one of the reasons that economies in developing countries grow so slowly is that they do not have well-developed financial markets. Does this argument make sense?

END OF QUESTIONS

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Formulas

Coupon Payment=

$$P = \frac{C}{(1+i)} + \frac{C}{(1+i)^2} + \frac{C}{(1+i)^3} + \dots + \frac{C}{(1+i)^n} + \frac{F}{(1+i)^n}$$

Expectation theory for n period bond=

$$i_{nt} = \frac{i_t + i_{t+1} + i_{t+2} + \dots + i_{t+(n-1)}}{n}$$

Treasury Bills discount rate=

$$i_{discount} = \frac{F - P}{F} \times \frac{360}{n}$$

Treasury bill investment rate=

$$i_{yt} = \frac{F - P}{P} \times \frac{365}{n}$$

Expected return= Sum(Prob × Price)

Variance= **Sum (Prob × (Price – Exp. Price)^2)**

Standard Deviation=$\sqrt{Prob(price - exp.price)^2}$

Liquidity Premium theory for n period bond=

$$i_{nt} = \frac{i_t + i^e_{t+1} + i^e_{t+2} + \dots + i^e_{t+(n-1)}}{n} + l_{nt}$$

Bond Rate of Return=

$$\frac{C + P_{t+1} - P_t}{P_t}$$

Pure Expectation theory for n period bond=

$$i_{nt} = \frac{i_t + i_{t+1} + i_{t+2} + \dots + i_{t+(n-1)}}{n}$$

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